Real-Time Secure Wallet Application

Project Document

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1. Project Title

Secure Digital Wallet Application with KYC Verification and Real-Time Transactions

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2. Project Overview

This project is a **real-time digital wallet system** that allows users to:

- Create an account and complete KYC verification
- Add money to a wallet
- Send and receive money instantly (P2P transfers)
- Make merchant payments
- Receive real-time transaction updates and notifications

The system is designed with **security, scalability, and compliance** in mind, similar to real-world applications like Paytm, PhonePe, or Google Pay.

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3. Problem Statement

Traditional payment systems are slow, not real-time, and lack proper tracking and transparency. Many systems also fail to meet security and KYC requirements.

Goal: Build a secure, real-time, and compliant wallet system that supports instant money transfers and proper identity verification.

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4. Objectives

- Provide instant wallet-to-wallet money transfers
- Ensure all users are verified using KYC
- Maintain accurate wallet balances using a ledger system
- Send real-time notifications for every transaction
- Secure user data and financial information

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5. Users of the System

- **End User** – Uses wallet for payments and transfers
- **Merchant** – Receives payments
- **Admin** – Manages KYC verification, disputes, and reports

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6. Functional Requirements

User Module

- User registration (email/phone + password)
- Login with JWT authentication
- Two-factor authentication (OTP)

KYC Module

- Upload ID proof and selfie
- Automatic KYC verification (OCR & face match)
- Manual verification by admin if required
- KYC status: PENDING, VERIFIED, REJECTED

Wallet Module

- Create wallet for each user
- Add money using payment gateway
- View wallet balance

Transaction Module

- Send money to another user
- Receive money instantly
- Maintain transaction history

Real-Time Notification Module

- Instant balance updates
- Real-time transaction alerts
- Push notifications / WebSocket updates

Admin Module

- Verify or reject KYC documents
- View transactions and user reports
- Handle disputes

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7. Non-Functional Requirements

- High security (encryption, authentication)
- High availability (24/7 access)
- Scalability (support many users)
- Fast response time (< 2 seconds)
- Data consistency and reliability

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8. System Architecture

Client (Mobile/Web App) ↓ API Gateway ↓ Backend Services

- Authentication Service
- User & KYC Service
- Wallet & Transaction Service
- Notification Service ↓ Database & Messaging
- PostgreSQL (User, Wallet, Transactions)
- Redis (Cache & Locks)
- Kafka (Real-Time Events)

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9. Technology Stack

Backend

- Java 17
- Spring Boot
- Spring Security (JWT)
- Hibernate / JPA

Frontend

- React / Angular (Web)
- Flutter / Android / iOS (Mobile)

Database

- PostgreSQL
- Redis

Messaging & Real-Time

- Apache Kafka
- WebSocket / Socket.IO

Cloud & DevOps

- Docker
- Kubernetes
- AWS (EC2, S3, RDS, IAM)

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10. Database Design

User Table

- user_id
- name
- email
- phone
- password_hash
- kyc_status

Wallet Table

- wallet_id
- user_id
- balance
- currency

Transaction Table

- transaction_id
- sender_wallet_id
- receiver_wallet_id
- amount
- status
- timestamp

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11. Real-Time Transaction Flow

User A sends ₹500 to User B

1. User A clicks “Send Money”
2. Request goes to backend with transaction details
3. System checks balance and KYC status
4. Amount is debited from User A wallet
5. Amount is credited to User B wallet
6. Transaction saved in database
7. Real-time notification sent to both users

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12. KYC Verification Flow

1. User uploads ID document and selfie
2. System performs OCR and face match
3. If successful → KYC VERIFIED
4. If mismatch → sent to admin for review
5. Admin approves or rejects KYC

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13. Security Features

- Password encryption (Argon2 / BCrypt)
- JWT-based authentication
- Role-based access control
- Data encryption at rest and in transit
- Secure API gateway with rate limiting

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14. Advantages of the System

- Instant money transfer
- High security and compliance
- Scalable microservices architecture
- Real-time user experience

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15. Future Enhancements

- QR-code based payments
- Card linking and tokenization
- AI-based fraud detection
- International payments

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16. Conclusion

This real-time secure wallet system provides a modern digital payment solution with strong security, KYC compliance, and instant transaction processing. It is suitable for real-world deployment and scalable for future growth.
